

EVIDENCE-GRADED · EDITION 01

10 Evidence-Backed Nootropic Stacks.

Ten combinations the literature actually supports — each graded by trial design, replication, and population, not by hype.

STACKS

10

Each with trial doses + Tier rating

CITATIONS

28

Inline · primary sources only

READING TIME

22 min

Designed for one focused sitting

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How we chose these stacks.

We pulled from ~800 studies in our research library and applied three filters before anything made this list.

01

Human evidence at the top of the stack

At least one ingredient with Tier-1 or Tier-2 human evidence — multiple RCTs, a meta-analysis, or a well-replicated single trial in the target population.

02

A coherent mechanism

Supporting ingredients need a mechanism that holds up — even when the human evidence is thinner. We say so out loud where it is.

03

A clean safety profile

At the dose ranges we recommend, with the contraindications named on each page. We never replace a clinician.

HOUSE RULE · STRUCTURE-AND-FUNCTION ONLY

Nootropics support cognitive function. They do not treat or cure conditions. If you suspect a clinical issue, talk to a clinician — that is not what this PDF is for. Effect sizes are reported the way the studies report them: actual deltas, the population studied, the duration. When evidence is mechanistic, animal-only, or single-small-study, we flag it with a hedge block.

EVIDENCE TIER LEGEND

TIER 1

Multiple RCTs or a meta-analysis. Replicated effect, named population.

TIER 2

One high-quality RCT, or multiple smaller trials with a coherent signal.

TIER 3

Preliminary, mechanistic, or animal-only. Italicized hedge body.

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Caffeine + L-Theanine

TARGET OUTCOME · Sharper attention without the jitter or crash.

01

TRIAL DOSE · PER SERVING

- 80–100 mg caffeine (≈ one cup of coffee)
- 200 mg L-theanine
- Ratio 1 : 2 caffeine to L-theanine

EVIDENCE

TIER 1

- Multiple RCTs in healthy adults
- 3 randomized double-blind trials
- Replicated on attention-switching tasks
- Acute effect, same-day onset

Of the dozen "focus" combinations marketed online, this is the one where the evidence and the marketing actually agree. In healthy adults, 200 mg of L-theanine paired with 100 mg of caffeine improved attention-switching accuracy and reduced susceptibility to distraction in a randomized crossover trial of 27 adults ¹. A second trial at a higher dose (150 mg + 250 mg) replicated the effect on simple reaction time and reduced "mental fatigue" ratings ², and a third trial in 44 healthy young adults extended the finding to task-switching accuracy and self-reported alertness ³.

Honest framing. The effect on attention is real and replicated, but modest — measurable in the lab on reaction-time and accuracy tasks, not a personality change. The "without jitter" benefit comes from L-theanine attenuating the rise in blood pressure and subjective anxiety some people get from caffeine.

WHO THIS IS FOR

Knowledge workers who already drink coffee and want a cleaner version of that effect.

WHO THIS IS NOT FOR

Caffeine-sensitive readers, anyone with hypertension, anyone pregnant, anyone using stimulant medication without clearing it with a clinician.

BUY

[L-Theanine 200mg on Amazon](#) · [Caffeine + L-Theanine combo on Amazon](#) tag · nootropicstk-

20

Bacopa monnieri

TARGET OUTCOME · Better memory recall and learning over weeks-to-months of consistent use.

02

TRIAL DOSE · PER DAY

- 300 mg extract, ≥ 50% bacosides
- Take with a meal containing fat
- Build over 8–12 weeks daily use

EVIDENCE

TIER 2 Replicated 12-week RCTs

- 3 RCTs, 12-week design
- Healthy adults + adults 55+
- Slow-build, not acute

Bacopa is the slow-builder on this list — its effect is on memory consolidation over months of daily use, not on how sharp you feel today. In 46 healthy adults, 300 mg/day standardized Bacopa for 12 weeks improved speed of visual information processing, learning rate, and memory consolidation⁴. In 48 community-dwelling adults aged 65+, the same dose for 12 weeks enhanced AVLT delayed word recall and reduced depression and anxiety scores⁵. A third 12-week RCT in 81 healthy adults over 55 replicated the verbal-learning and delayed-recall improvements on the Rey AVLT⁶.

Honest framing. Bacopa improved verbal memory recall on the order of a 9–12% delta over placebo across these trials. That is a real, replicated effect — and it is not a "limitless pill." It is a quiet, slow-building improvement on memory tasks in adults who took it daily for 3 months.

PRELIMINARY EVIDENCE

GI side effects (loose stools, mild nausea) are common at the start. Take with food and consider splitting the dose AM/PM for the first two weeks.

WHO THIS IS FOR

Students, researchers, anyone preparing for a sustained learning project who can commit to 8–12 weeks.

WHO THIS IS NOT FOR

Anyone wanting an acute "today" effect.
Anyone on thyroid medication (animal data suggests possible interaction — clear with your clinician).

BUY

[Bacopa monnieri 50% bacosides on Amazon](#) tag · nootropicstk-20

Lion's Mane (*Hericium erinaceus*)

TARGET OUTCOME · Mild cognitive support, particularly in older adults.

03

TRIAL DOSE · PER DAY

- 1,000 mg fruiting-body extract
- Verified beta-glucan ≥ 25%
- Split AM / midday

EVIDENCE

TIER 3 Small trials in older adults

- 2 RCTs in adults 50–80 with MCI
- n = 30 (Mori), n > 50 (Saitzu)
- Healthy-adult evidence is thin

In 30 Japanese adults aged 50–80 with mild cognitive impairment, 3 g/day of *Hericium erinaceus* for 16 weeks improved cognitive function scores (HDS-R) — though the benefit faded 4 weeks after discontinuation ⁷. A 12-week double-blind RCT in older adults reported significant improvements on MMSE scores and prevented further deterioration ⁸.

PRELIMINARY EVIDENCE

The Mori 2009 trial is the most-cited piece of evidence here, and it was small (n = 30) and in adults with mild cognitive impairment, not healthy young adults. The "Lion's Mane regrows your brain" claim you see on TikTok is overreach — the mechanism (NGF stimulation) is supported in cell and animal work, but human evidence in healthy adults is thin. We include it because the evidence we do have is consistent and the safety profile is clean.

Avoid mycelium-on-grain products that don't separate the mushroom from the substrate — beta-glucan content drops, and you're often paying for grain.

WHO THIS IS FOR

Older adults looking for cognitive support; readers who want to try a mushroom adaptogen with the most human data of any of them.

WHO THIS IS NOT FOR

Anyone expecting a same-day acute effect.
Anyone with a known mushroom allergy.

BUY

[Lion's Mane fruiting-body extract on Amazon](#) tag · nootropicstk-20

Rhodiola rosea

TARGET OUTCOME · Reduced fatigue and improved mental performance under stress.

04

TRIAL DOSE · PER DAY

- 200–400 mg standardized extract
- 3% rosavins / 1% salidroside
- AM, on an empty stomach

EVIDENCE

TIER 2

Replicated trials, fatigue under stress

- 2 placebo-controlled RCTs (SHR-5)
- 1 open-label supportive trial
- Acute + 4-week effects

In 60 adults aged 20–55 with stress-related fatigue, 576 mg/day of SHR-5 extract for 28 days reduced burnout-related fatigue and improved attention⁹. A 6-week double-blind trial in 89 adults with mild-to-moderate depression found that SHR-5 at 340 or 680 mg/day reduced overall depression, insomnia, emotional instability, and somatization vs. placebo¹⁰. A separate open-label trial in 101 adults with life-stress symptoms reported clinically relevant reductions in stress symptoms and functional impairment over 4 weeks¹¹.

Honest framing. The strongest evidence is for **fatigue reduction under stress**, not raw cognitive enhancement in unstressed people. Effect sizes are moderate. The Edwards 2012 trial is open-label (no placebo arm) — useful as supportive evidence but not headline.

WHO THIS IS FOR

People in a high-demand work or study sprint, or recovering from burnout-adjacent fatigue.

WHO THIS IS NOT FOR

Anyone on antidepressants (potential serotonergic interaction — clear with your clinician). Anyone with bipolar disorder (case reports of activation).

BUY

[Rhodiola rosea 3% rosavins on Amazon](#) tag · nootropicstk-20

Creatine monohydrate

TARGET OUTCOME · Improved cognitive performance under sleep deprivation, mental fatigue, or low energy availability.

05

TRIAL DOSE · PER DAY

- 5 g creatine monohydrate
- Daily, with or without food
- No loading phase needed

EVIDENCE

TIER 1 Systematic review, 6 RCTs

- Meta-analysis n = 281
- Strongest under metabolic stress
- Decades of safety data

Creatine is the best-studied performance supplement on the planet, and the cognitive evidence has caught up. In 45 healthy young vegetarian adults, 5 g/day creatine for 6 weeks improved working memory (backward digit span) and intelligence on the Raven's Advanced Progressive Matrices¹². In sleep-deprived participants, 7-day creatine pre-loading attenuated central-executive task decrements at 36 hours of sleep deprivation with intermittent moderate exercise¹³. A systematic review of 6 RCTs (n = 281) concluded short-term memory and intelligence/reasoning may be improved by creatine in healthy individuals¹⁴.

Honest framing. Creatine's cognitive benefits are most pronounced under metabolic stress (sleep deprivation, vegetarian/vegan diet, demanding cognitive load). In well-rested, omnivorous, low-stress conditions the effect is smaller and harder to feel. The headline is *"more resilient when conditions are bad,"* not *"smarter when conditions are good."*

WHO THIS IS FOR

Students pulling long stretches, anyone on a vegetarian/vegan diet (lower baseline creatine), anyone who trains hard.

WHO THIS IS NOT FOR

People with kidney disease (talk to a clinician). Note: creatine causes ~1–2 lb of water-weight gain — that is not fat.

BUY

[Creatine monohydrate \(micronized\) on Amazon](#) tag · nootropicstk-20

Omega-3 EPA / DHA

TARGET OUTCOME · Cognitive support, mood support, and a structural foundation for the brain — over months and years.

06

TRIAL DOSE · PER DAY

- 1,000–2,000 mg EPA + DHA combined
- Fish oil or algal oil
- Third-party tested · TOTOX < 10

EVIDENCE

TIER 1 Multiple RCTs across populations

- MIDAS trial · n = 485 adults 55+
- n = 176 healthy 18–45
- Replicates on memory + executive function

The MIDAS trial — 485 adults aged 55+ with age-related cognitive decline — found that 900 mg/day DHA for 24 weeks improved Paired Associates Learning errors and Verbal Recognition Memory¹⁵. In 176 healthy adults aged 18–45, 1.16 g/day DHA for 6 months improved reaction times of episodic and working memory, with episodic-memory accuracy gains significant in women¹⁶. A 26-week RCT in 65 healthy adults aged 50–75 found that 2.2 g/day long-chain omega-3 improved executive function and white-matter microstructural integrity¹⁷.

Honest framing. Omega-3s are foundational, not flashy. The cognitive deltas are small per study, but they replicate across populations and outcomes, and they compound over time. If your diet has fewer than 2 servings of fatty fish per week, this is the highest-leverage supplement in this PDF.

WHO THIS IS FOR

Almost everyone — but especially anyone with low fish intake, vegetarian or vegan readers (consider algal oil for DHA), and people over 50.

WHO THIS IS NOT FOR

Anyone on blood thinners (clear high doses with your clinician). Anyone with a fish or shellfish allergy — use algal oil instead.

BUY

[Triple-strength fish oil \(EPA + DHA\) on Amazon](#) · [Algal oil DHA on Amazon](#) tag · nootropicstk-

20

Ashwagandha (*Withania somnifera*)

TARGET OUTCOME · Reduced perceived stress and lower cortisol over 4–8 weeks.

07

TRIAL DOSE · PER DAY

- 240–600 mg root extract
- KSM-66, Sensoril, or Shoden
- With food · build over 4 weeks

EVIDENCE

TIER 1

Replicated across branded extracts

- 3 RCTs on stress + cortisol
- KSM-66, generic, and Shoden tested
- Self-report + serum cortisol outcomes

In 64 chronically stressed adults, 300 mg of high-concentration KSM-66 root extract twice daily for 60 days produced significant reductions across all stress-assessment scales ($P < 0.0001$, ~44% reduction in Perceived Stress Scale scores) and serum cortisol ($P = 0.0006$, ~28% reduction) vs. placebo¹⁸. In 60 healthy adults, 250 mg/day or 600 mg/day *Withania somnifera* for 8 weeks reduced PSS scores and serum cortisol vs. placebo¹⁹. A separate 60-day trial in 60 self-reported high-stress adults using 240 mg/day Shoden extract reduced HAM-A anxiety scores and morning cortisol vs. placebo²⁰.

Honest framing. The cortisol reduction is real and replicated, across multiple branded extracts. The "lower stress score" is a self-report measure — useful, but not the same as a behavioral or physiological outcome. Effect sizes were moderate-to-large, but populations were specifically *chronically stressed* adults — a healthier baseline will likely show smaller benefits.

WHO THIS IS FOR

Adults with sustained life stress; readers who want to support sleep onset (mildly sedating for some).

WHO THIS IS NOT FOR

Pregnant or breastfeeding people. Anyone with hyperthyroidism or on thyroid medication. Anyone with autoimmune conditions on immunosuppressants.

BUY

[Ashwagandha KSM-66 600mg on Amazon](#) tag · nootropicstk-20

L-Tyrosine

TARGET OUTCOME · Maintained cognitive performance during acute stress, sleep loss, or cold exposure.

08

TRIAL DOSE · PER DOSE

- Trial range 100–150 mg/kg bodyweight
- Practical clinical dose 2 g
- 30–60 min before stressor · empty

EVIDENCE

TIER 2

Situational, replicated under stress

- 3 RCTs · cold, sleep loss, switching
- Acute, situational tool
- Body adapts if taken daily

Under cold-water immersion, 150 mg/kg L-tyrosine increased correct responses on a Match-to-Sample working-memory task vs. placebo²¹. In 22 healthy adults under a double-blind protocol, L-tyrosine reduced task-switching costs (improved cognitive flexibility) vs. placebo²². Across ~13 hours of extended wakefulness, 150 mg/kg L-tyrosine in split doses attenuated the usual decline on a psychomotor task and reduced lapse probability on a high-event-rate vigilance task²³.

Honest framing. L-Tyrosine helps the brain replenish dopamine and norepinephrine when those neurotransmitters are being depleted by acute stress. Translation: it does *not* make a well-rested, calm person sharper — it helps a stressed person stay closer to baseline. It is a situational tool, not a daily one. Do not take with high-protein meals (competing amino acids reduce absorption).

WHO THIS IS FOR

Pre-exam, pre-presentation, post-poor-sleep, jet-lagged, cold-exposed (literally).

WHO THIS IS NOT FOR

Anyone on MAO inhibitors. Anyone with hyperthyroidism. Anyone with melanoma history (tyrosine is a melanin precursor — talk to a clinician). Not effective if taken daily; the body adapts.

BUY

[L-Tyrosine 500mg on Amazon](#) tag · nootropicstk-20

Citicoline (*CDP-Choline*)

TARGET OUTCOME · Improved attention, processing speed, and memory consolidation.

09

TRIAL DOSE · PER DAY

- 250–500 mg citicoline (Cognizin)
- With breakfast
- 4-week+ build for healthy adults

EVIDENCE

TIER 3

Thin in healthy adults; clean in clinical pop

- 1 RCT in adolescents
- 1 RCT post-stroke (older adults)
- Healthy-adult RCTs are limited

In 75 healthy adolescent males, 250 or 500 mg/day citicoline for 28 days improved attention and psychomotor speed and decreased impulsivity vs. placebo²⁴. In a much larger trial of 347 first-ever ischemic-stroke patients, 1 g/day citicoline for 12 months improved attention-executive function and temporal orientation vs. control²⁵.

PRELIMINARY EVIDENCE

Citicoline is a precursor to phosphatidylcholine and a source of choline for acetylcholine production. The healthy-population evidence is thinner than the marketing suggests — the cleanest signal we have is in adolescent males; large adult-only RCTs are limited. The strongest evidence is in clinical populations (post-stroke vascular cognitive impairment) — relevant for older readers, less directly translatable to healthy 25-year-olds.

WHO THIS IS FOR

Older adults, anyone on a low-egg / low-meat diet (often low on choline), anyone running a stack with a cholinergic component (e.g., racetams — beyond this PDF's scope).

WHO THIS IS NOT FOR

Anyone with bipolar disorder (case reports of activation). Anyone who has taken citicoline and felt agitated — it is cholinergic, and not everyone tolerates it.

BUY

[Cognizin Citicoline 250mg on Amazon](#) tag · nootropicstk-20

Phosphatidylserine

TARGET OUTCOME · Memory and learning support, particularly in older adults.

10

TRIAL DOSE · PER DAY

- 300 mg total · 100 mg × 3
- Soy- or sunflower-derived
- With meals containing fat

EVIDENCE

TIER 2 Replicated in older adults

- 3 RCTs · 12 weeks–6 months
- Adults with memory complaints
- Modern soy/sunflower not bovine

In 157 non-demented older adults with memory complaints, a 15-week trial of phosphatidylserine combined with omega-3 (PS-DHA) significantly improved verbal immediate recall, with gains in immediate and delayed recall, learning, and processing speed strongest in those with higher baseline cognition²⁶. In elderly Japanese subjects with memory complaints, 100 mg or 300 mg/day soybean-derived phosphatidylserine for 6 months improved memory functions, with delayed-verbal-recall gains strongest in those with lower baseline scores²⁷. In 149 adults with age-associated memory impairment, 100 mg three times daily (300 mg/day) for 12 weeks improved learning and memory tasks of daily life vs. placebo²⁸.

PRELIMINARY EVIDENCE

The memory-improvement evidence is strongest in older adults with subjective memory complaints, not healthy young adults. Some older PS trials (Crook 1991) used bovine-derived PS, which is no longer commercially available (BSE / mad-cow concerns); modern products use soy or sunflower lecithin, and we don't have a head-to-head efficacy comparison. PS is also widely marketed for cortisol reduction during exercise — that evidence is weaker and we are not making that claim here.

WHO THIS IS FOR

Adults 50+ who want a memory-focused stack with reasonable human evidence, often paired with omega-3s.

WHO THIS IS NOT FOR

Anyone with a soy allergy (use sunflower-derived). Anyone expecting a same-day effect.

BUY

[Phosphatidylserine 100mg on Amazon](#) tag · nootropicstk-20

How to actually use this PDF.

- 1 **Pick one stack at a time.** Do not start three at once — you won't know what is working.
- 2 **Give it the time the studies gave it.** Bacopa needs 8–12 weeks. Ashwagandha needs 4–8. Caffeine + L-theanine works the same day. Match your expectations to the timeline.
- 3 **Track one outcome.** Memory recall, sleep onset, perceived stress at 5 PM — pick one thing and rate it daily on a 1–10 scale. Real signal beats vibes.
- 4 **If a stack isn't working at the duration the study used, stop.** A nootropic that doesn't work for *you* isn't a moral failing. Stop and try the next one.

What we deliberately left off.

We left off **modafinil** and **racetams** (prescription / grey-market, beyond an introductory list); **nicotine** (effective at low doses, but the addictive-substance disclosure and harm-reduction conversation deserves its own piece); **psilocybin microdosing** (the human RCT evidence on cognitive endpoints is weaker than the hype suggests); and **noopept** and **phenylpiracetam** (regulatory grey area in the US, evidence base is thinner than the marketing).

If we cited it loudly, we believed it. If we left it off, we didn't.

Sources & disclosure.

- Every claim in this PDF is linked to a PubMed citation. If you find a dead link or a citation that doesn't support the claim, write to us at info@nootropicstacker.com and we will fix it in the next revision.
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VH

Vera Huang, CMO · NootropicStacker

Editorial review by NootropicStacker editorial board.

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